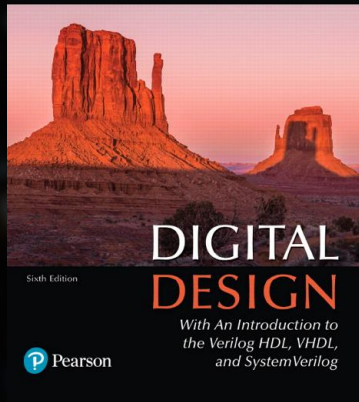
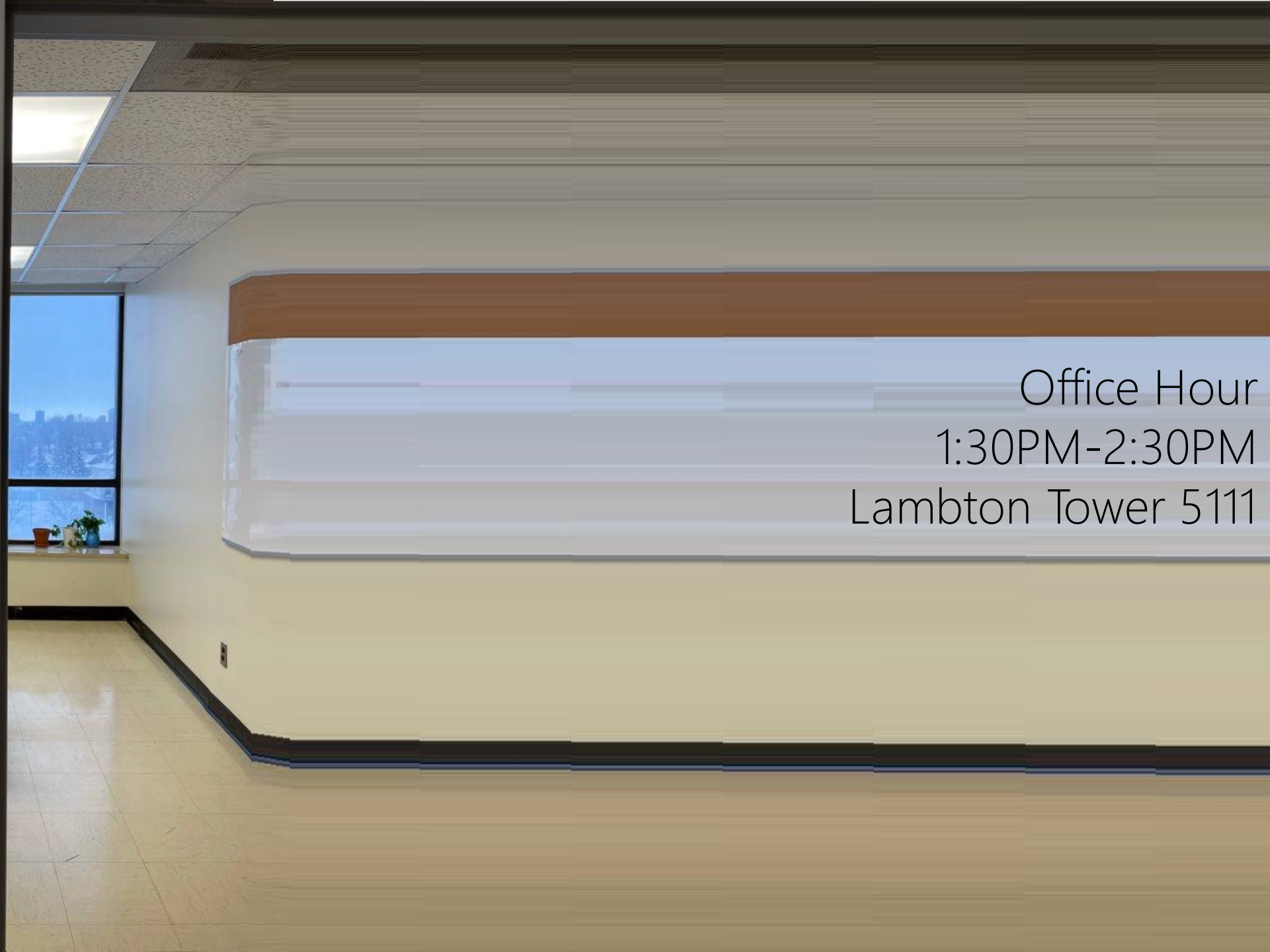


M. Morris Mano • Michael D. Ciletti

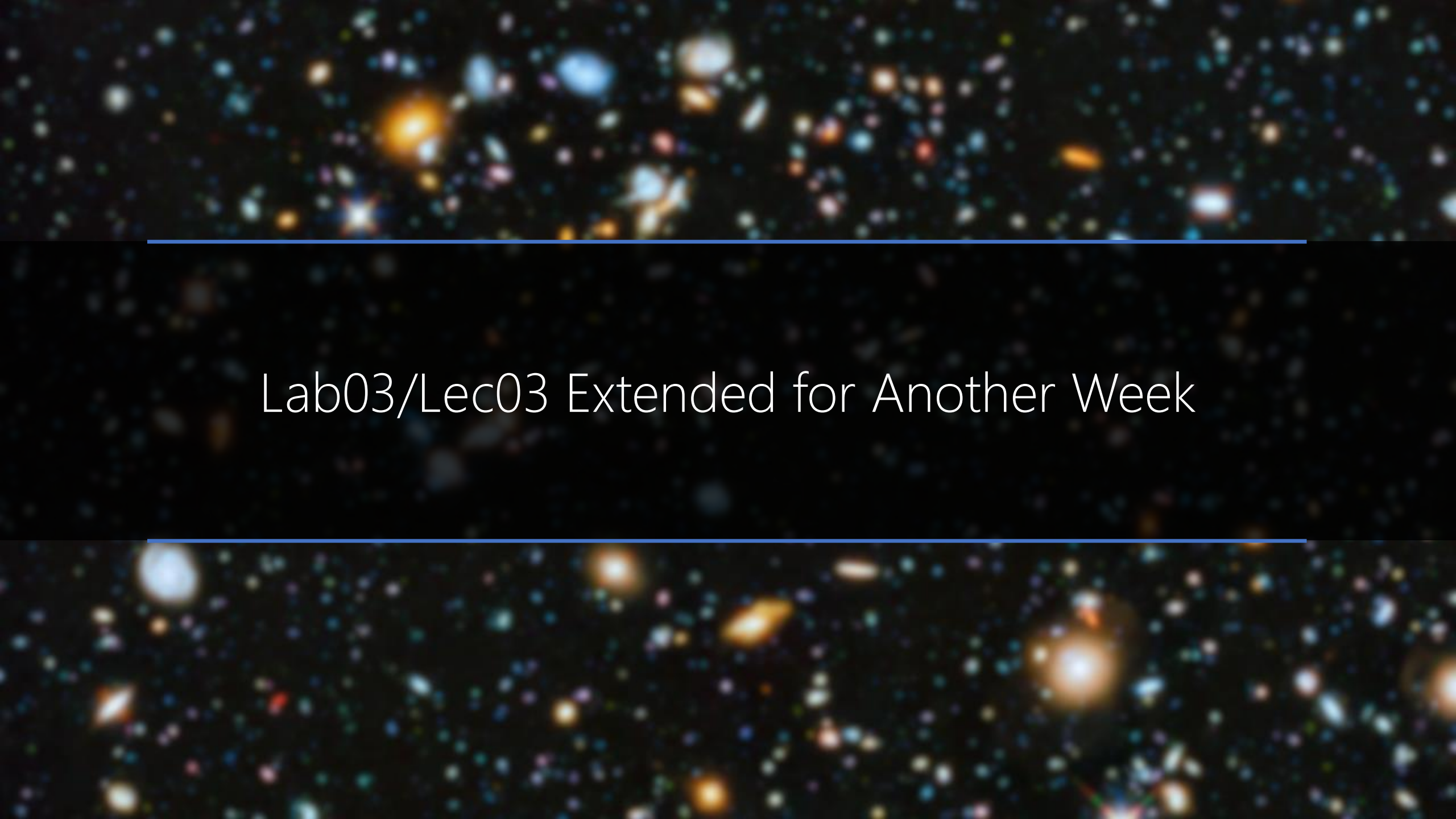


Chapter 1

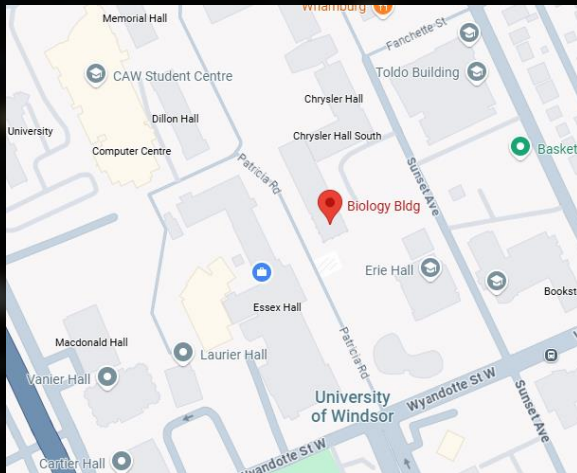
Digital Systems and Binary Numbers



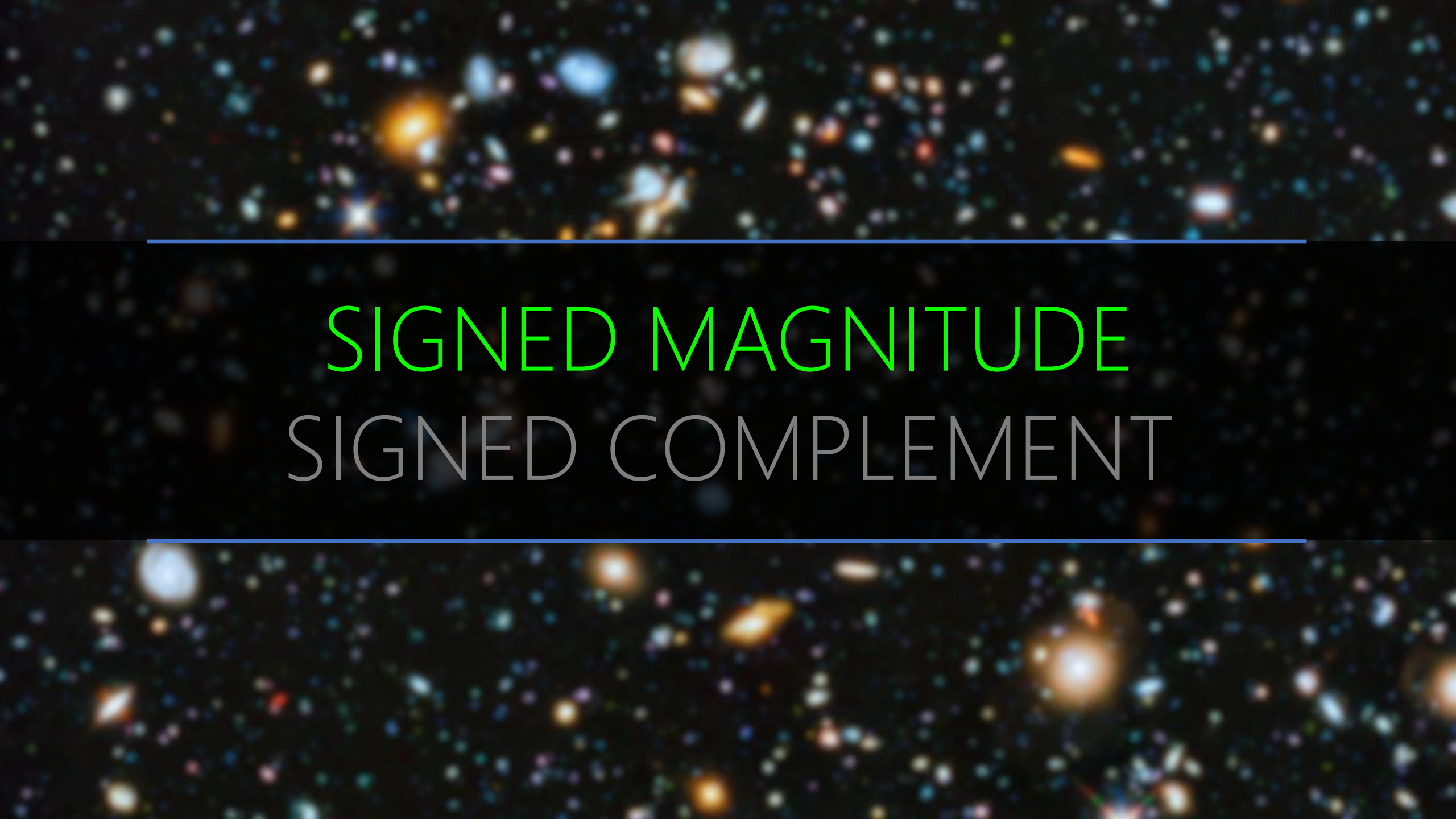
Office Hour
1:30PM-2:30PM
Lambton Tower 5111

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

Lab03/Lec03 Extended for Another Week



Final Exam (Tentative)
12-Aug, 8:30 AM
Biology Building (BB) 113

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots in various shapes and sizes, including spirals, ellipses, and irregular forms, set against a dark, star-filled sky. Two thin, horizontal blue lines are positioned above and below the central text.

SIGNED MAGNITUDE

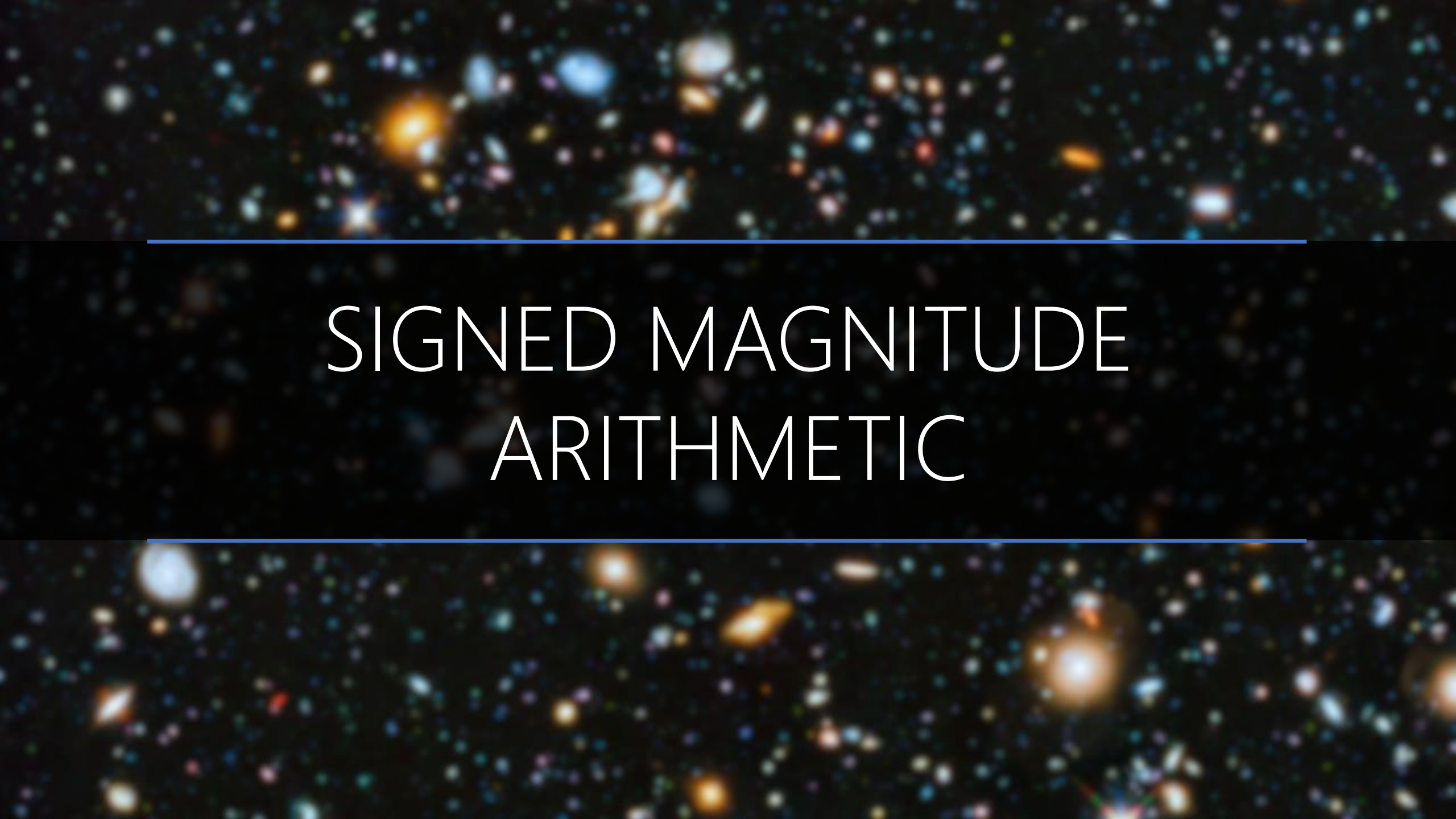
SIGNED COMPLEMENT

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					



Signed

Magnitude

The background of the slide is a deep space image showing a vast field of galaxies and stars. The galaxies are in various stages of evolution, some appearing as bright, irregular blobs of light, while others are more structured. The stars are small, distant points of light, many of which are multi-colored, suggesting they have been filtered through different wavelength bands. The overall scene is a dense, colorful mosaic of celestial objects against the blackness of space.

SIGNED MAGNITUDE ARITHMETIC

0	X
0	X
1	X

+

0	Y
1	Y
1	Y

=

0	$X+Y$
$X < Y$ (if borrow)	$X-Y$ (if borrow, apply it)
1	$-(X+Y) = X+Y$

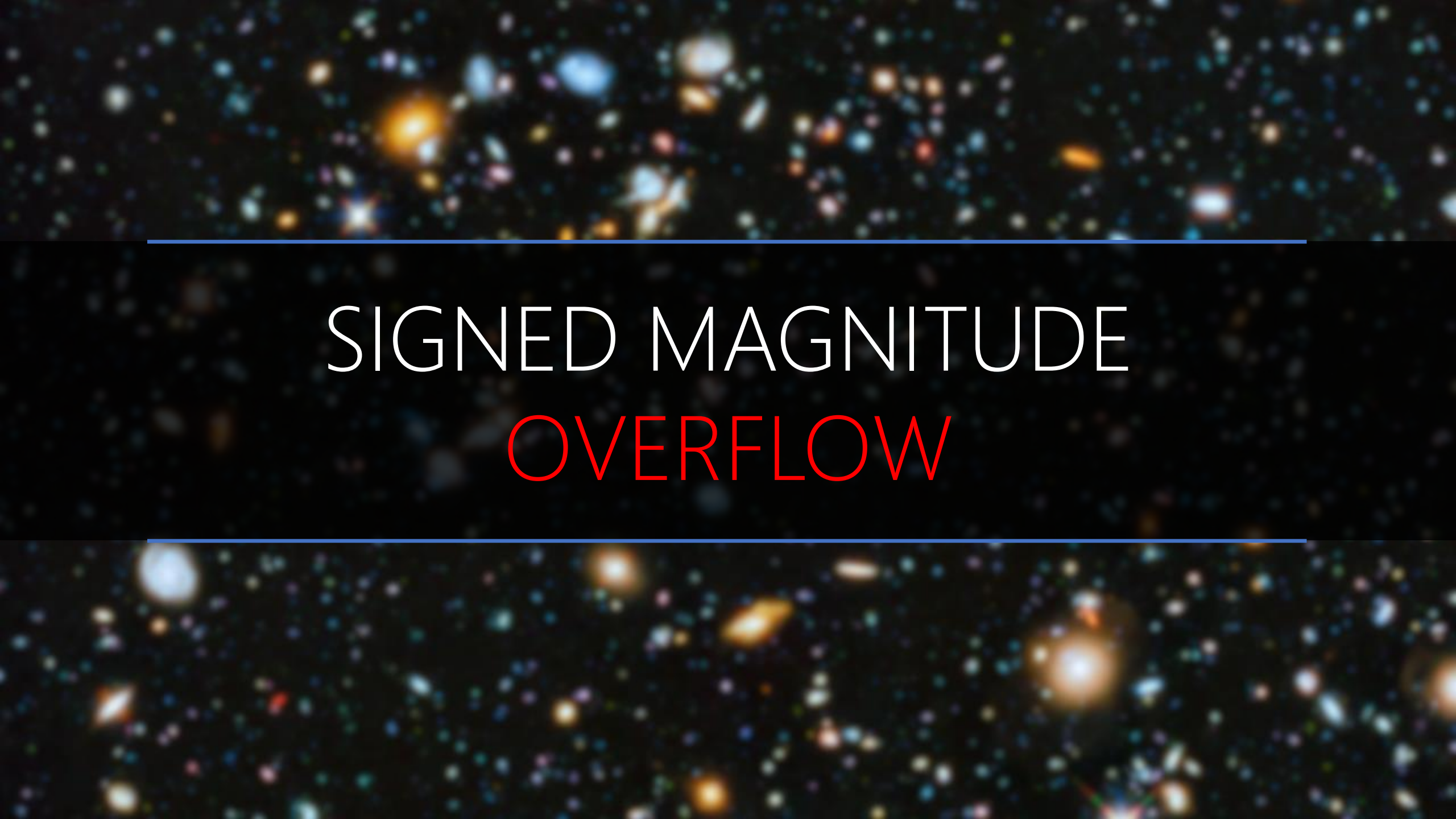
0	X
0	X
1	X

-

0	Y
1	Y
1	Y

=

$X < Y$ (if borrow)	$X-Y$ (if borrow, apply it)
0	$X+Y$
$X > Y$ (if borrow)	$-X+Y = Y-X$ (if borrow, apply it)

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots in various shapes and sizes, including spirals, ellipticals, and irregular forms, set against a dark, star-filled sky. Two thin, horizontal blue lines are positioned above and below the central text.

SIGNED MAGNITUDE OVERFLOW

Overflow!

0	X
0	X
1	X

+

0	Y
1	Y
1	Y

=

0 → 1	X+Y
X<Y	X-Y
1 → 1	X+Y

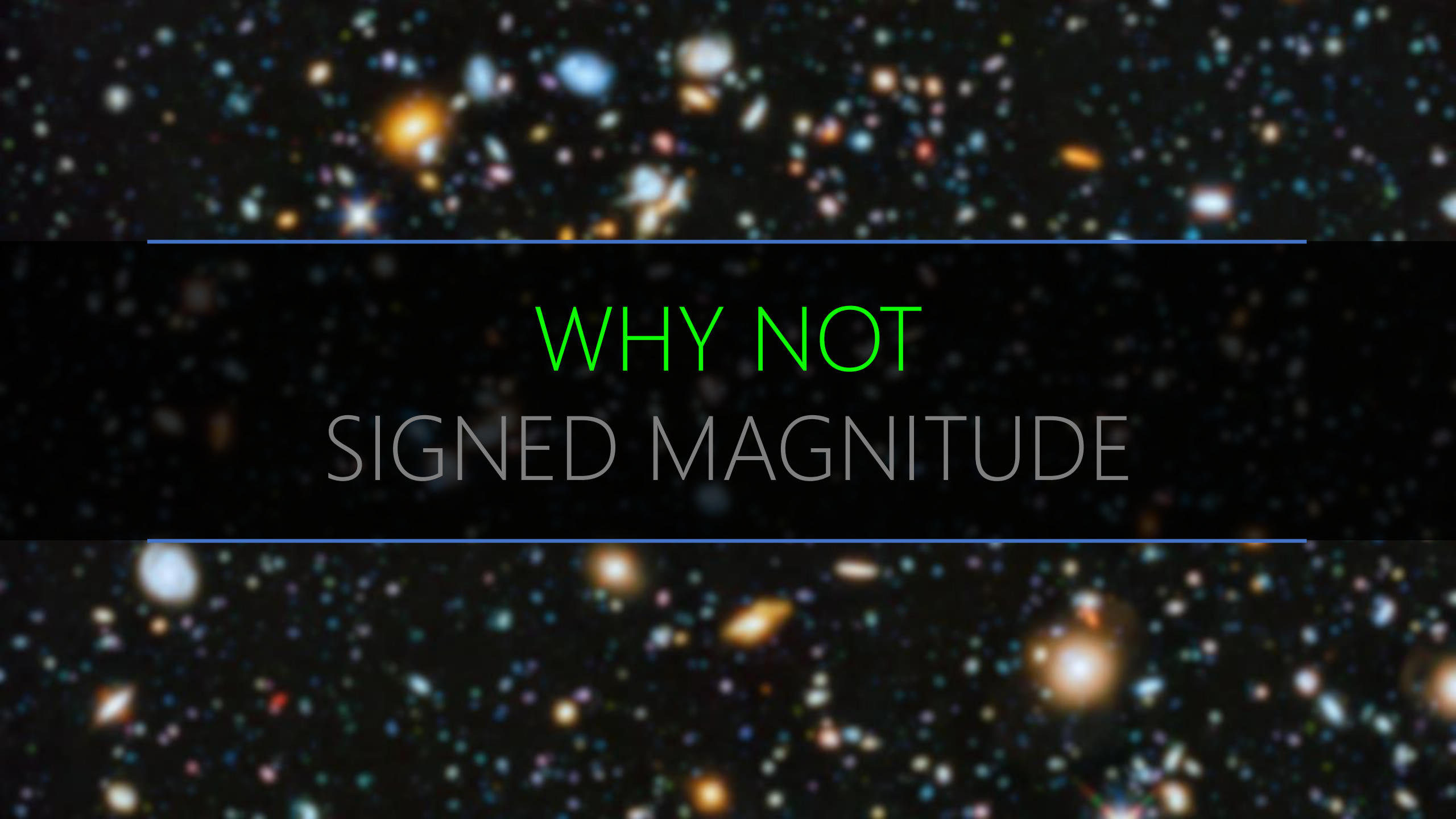
0	X
0	X
1	X

-

0	Y
1	Y
1	Y

=

X<Y	X-Y
0 → 1	X+Y
X>Y	-X+Y=Y-X

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots in various shapes and sizes, including spirals, ellipticals, and irregular forms, set against a dark, star-filled sky. Two thin, horizontal blue lines are positioned above and below the central text, framing it.

WHY NOT SIGNED MAGNITUDE

Give up left most position for sign! What are the wastes?

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$+0 \rightarrow \text{Max} = r^{n-1} - 1 = r^n - 1$$

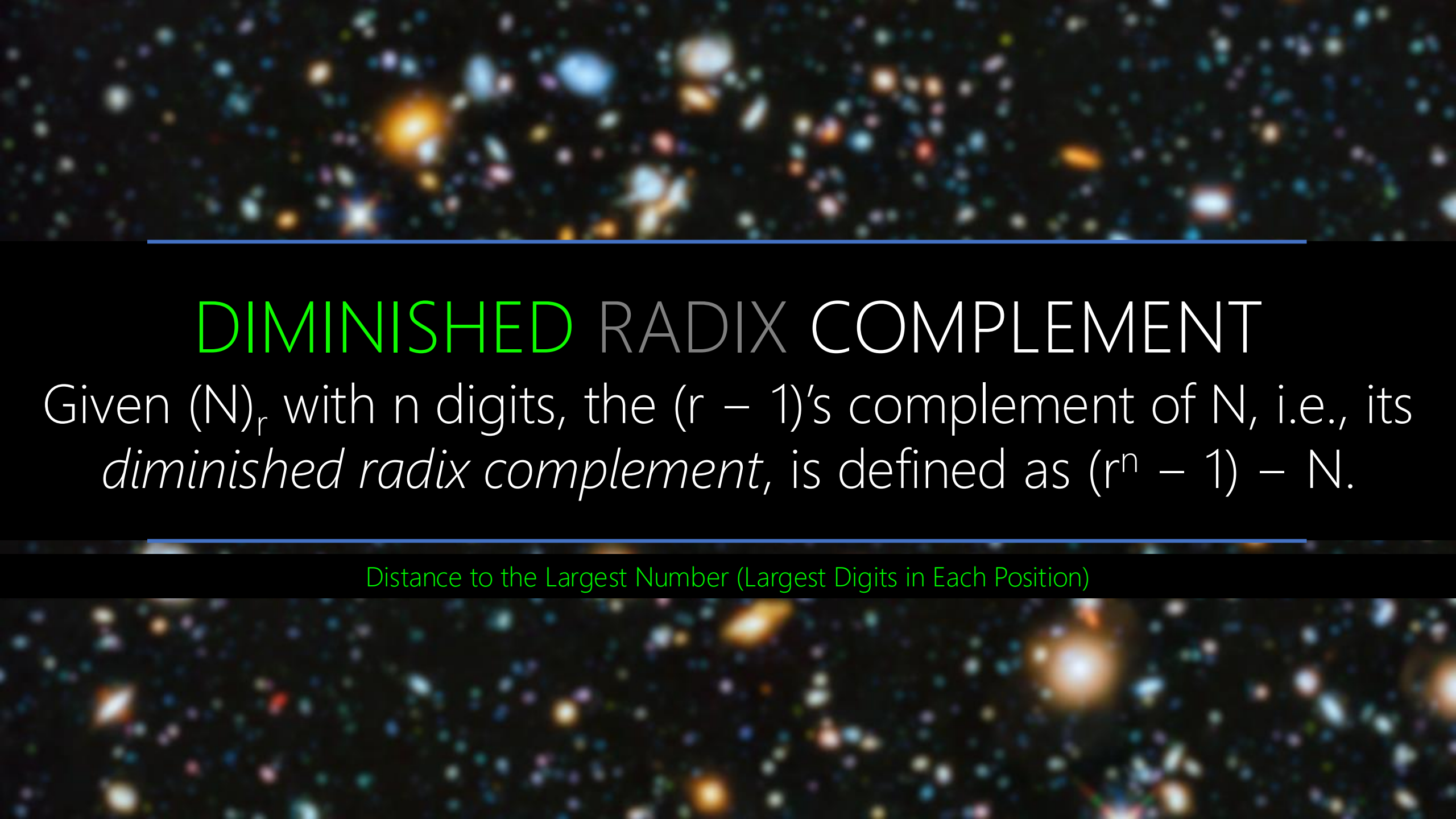
$$\text{Min} = -(r^{n-1} - 1) \leftarrow -0$$

A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in shades of blue, orange, and white, set against a dark, star-filled background. Two horizontal blue lines are positioned above and below the central text.

SIGNED COMPLEMENT

A deep space image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

SIGNED COMPLEMENT

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. A horizontal blue line is positioned above the title.

DIMINISHED RADIX COMPLEMENT

Given $(N)_r$ with n digits, the $(r - 1)$'s complement of N , i.e., its *diminished radix complement*, is defined as $(r^n - 1) - N$.

Distance to the Largest Number (Largest Digits in Each Position)

A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various colors, including blue, orange, and white, and are scattered across a dark background. Two horizontal blue lines are drawn across the image, one above and one below the central text.

1's COMP. BASE-2

Base-2		2^4	2^3	2^2	2^1	2^0
$2^5 =$	1	0	0	0	0	0

Base-2		2^4	2^3	2^2	2^1	2^0
$2^5 =$	1	0	0	0	0	0
$1 =$		0	0	0	0	1

Base-2		2^4	2^3	2^2	2^1	2^0
		+2	+2	+2	+2	
	-1	-1	-1	-1	-1	+2
$2^5 =$	1	0	0	0	0	0
$1 =$		0	0	0	0	1
$2^5 - 1 =$		1	1	1	1	1
	5 digits of 1					

Base-2	2^5	2^4	2^3	2^2	2^1	2^0
$2^5-1=$		1	1	1	1	1
N=		1	0	1	0	1
$(2^5-1)-N=$		0	1	0	1	0

1's complement of $(10101)_2 = (01010)_2 =$ NOT on each digit

A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in a variety of hues, including blue, orange, yellow, and white, set against a dark, star-filled background. Two horizontal blue lines are positioned above and below the central text.

3's COMP. BASE-4

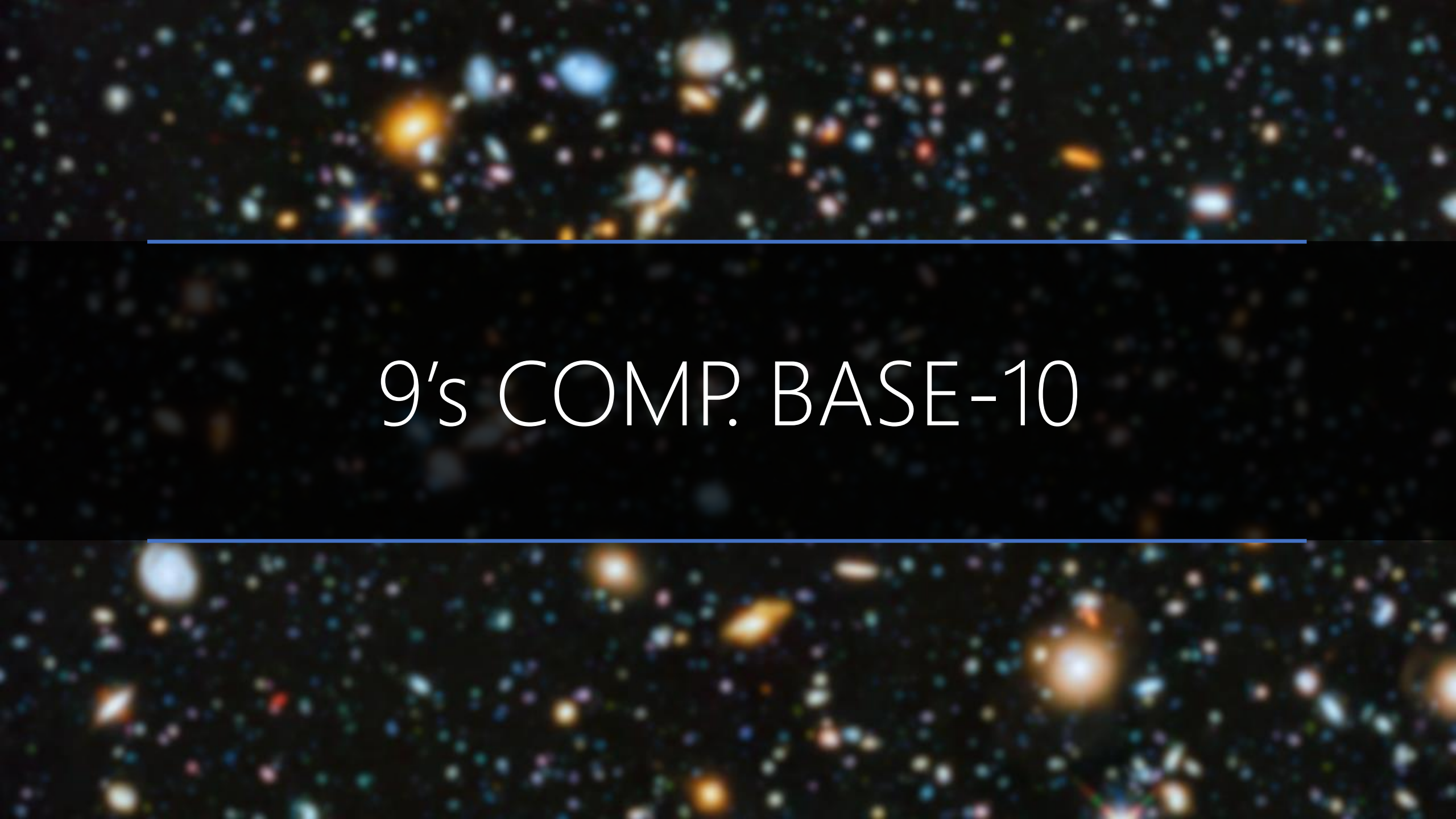
Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 =$	1	0	0	0	0	0

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 =$	1	0	0	0	0	0
$1 =$	0	0	0	0	0	1

Base-4		4^4	4^3	4^2	4^1	4^0
		$+4$	$+4$	$+4$	$+4$	
	-1	-1	-1	-1	-1	$+4$
$4^5 =$	1	0	0	0	0	0
1 =		0	0	0	0	1
$4^5 - 1 =$		3	3	3	3	3
	5 digits of 3					

Base-4		4^4	4^3	4^2	4^1	4^0
$4^5 - 1 =$		3	3	3	3	3
N =		1	2	1	3	0
$(4^5 - 1) - N =$		2	1	2	0	3

3's complement of $(12130)_4 = (21203)_4 = 3 - \text{Each digit}$

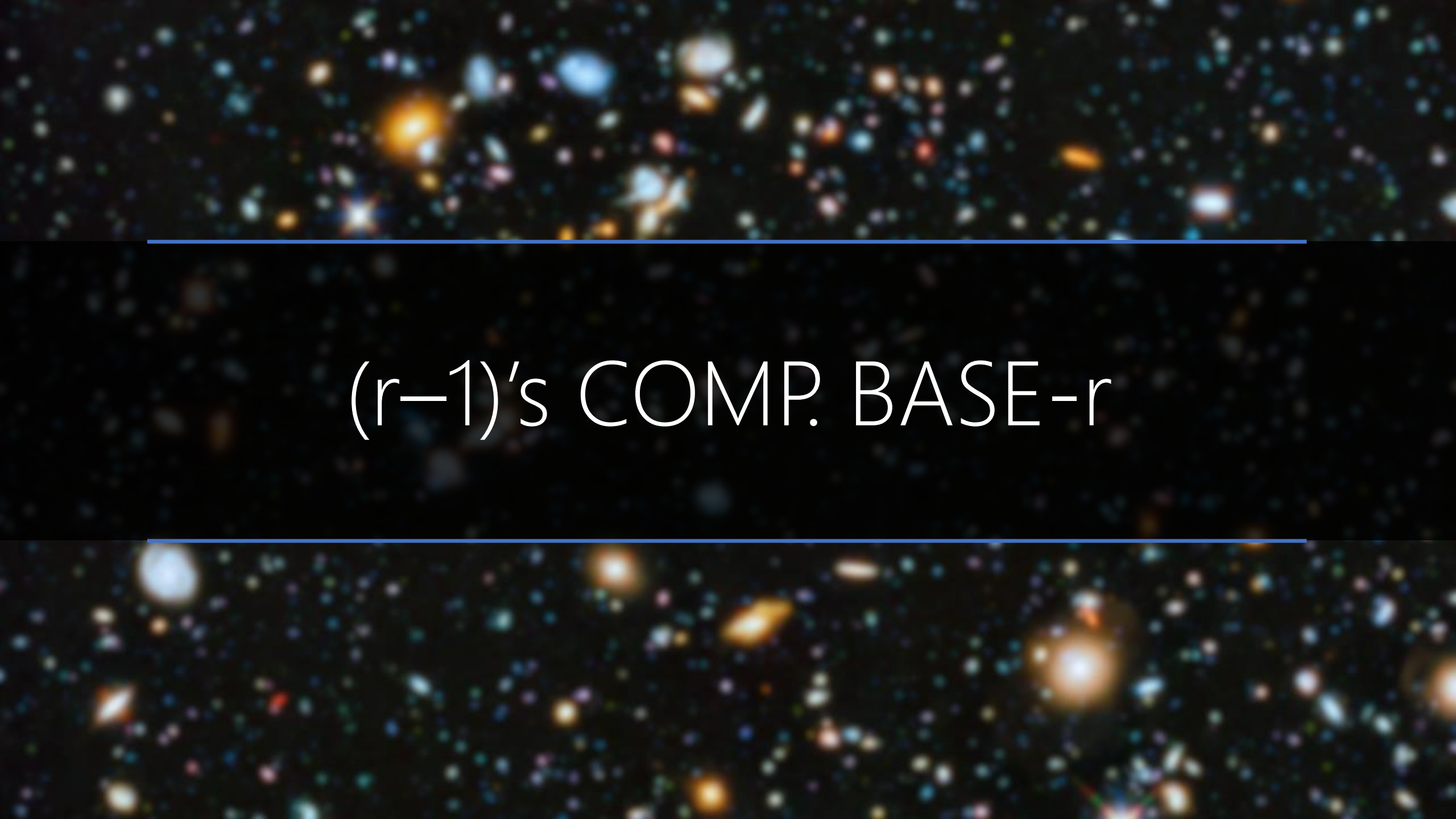
A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in a variety of hues, including blue, orange, yellow, and white, set against a dark, black background. The galaxies are distributed across the entire frame, with some appearing more prominent than others.

9's COMP. BASE-10

Base-10		10^4	10^3	10^2	10^1	10^0
		+10	+10	+10	+10	
	-1	-1	-1	-1	-1	+10
$10^5 =$	1	0	0	0	0	0
$1 =$		0	0	0	0	1
$10^5 - 1 =$		9	9	9	9	9
	5 digits of 9					

Base-10		10^4	10^3	10^2	10^1	10^0
$10^5 - 1 =$		9	9	9	9	9
$N =$		1	2	1	3	0
$(10^5 - 1) - N =$		8	7	8	6	9

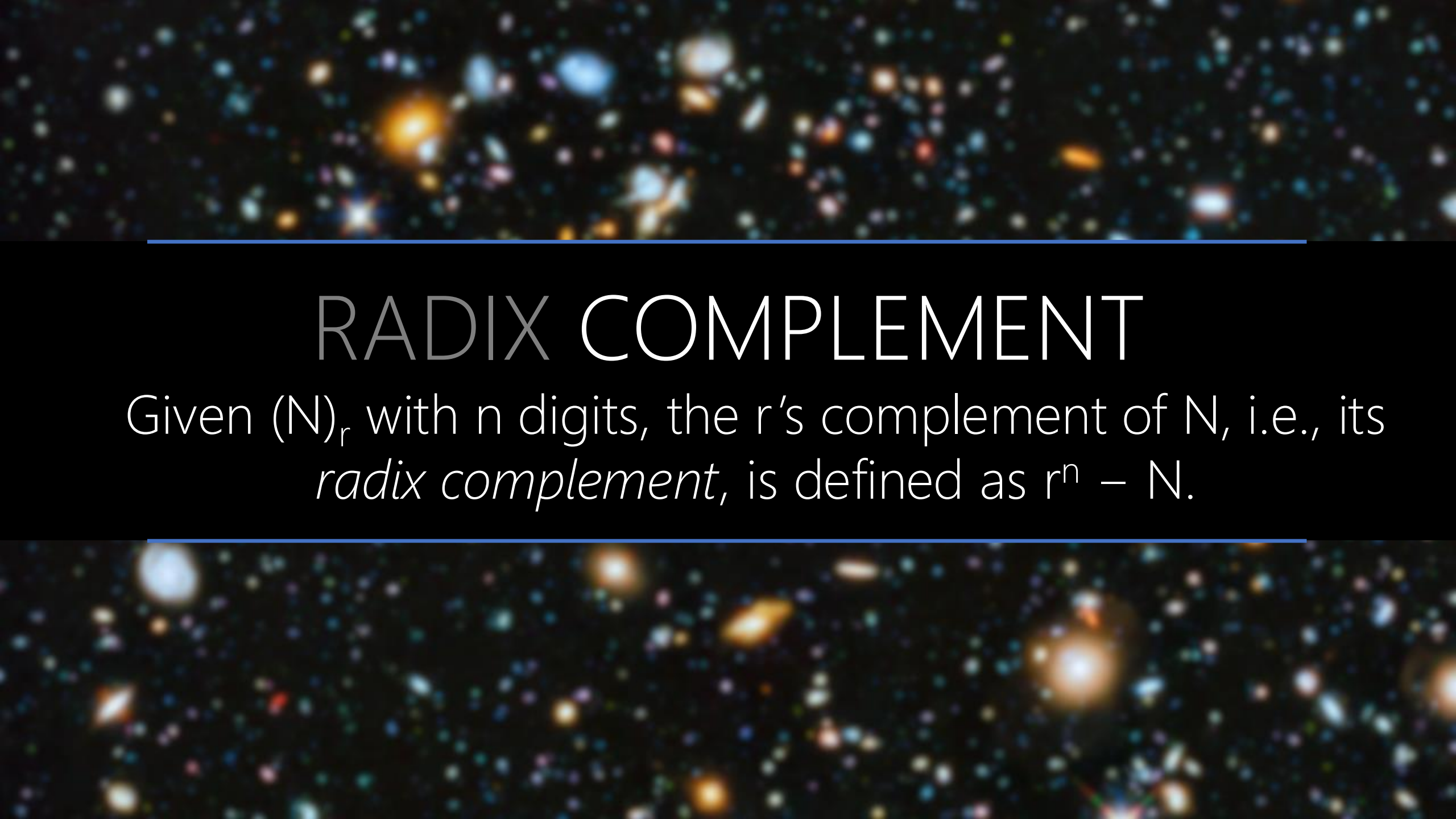
9's complement of $(12130)_{10} = (87869)_{10} = 9 - \text{Each digit}$

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

$(r-1)$'s COMP. BASE- r

Base-r		r^{n-1}	...	r^2	r^1	r^0
$r^n-1=$		$r-1$	...	$r-1$	$r-1$	$r-1$
$N=$		d_{n-1}	...	d_2	d_1	d_0
$(r^n-1)-N=$		$r-1-d_{n-1}$	...	$r-1-d_{n-1}$	$r-1-d_{n-1}$	$r-1-d_{n-1}$

$(r-1)$'s complement of $(N)_r = (r-1) - \text{Each digit}$



RADIX COMPLEMENT

Given $(N)_r$ with n digits, the r 's complement of N , i.e., its *radix complement*, is defined as $r^n - N$.

RADIX COMPLEMENT

Given $(N)_r$ with n digits, the r 's complement of N , i.e., its *radix complement*, is defined as $r^n - N$.

Diminished Complement + 1

$$(r-1)\text{'s Complement} + 1 = [(r^n - 1) - N] + 1 = r^n - N$$

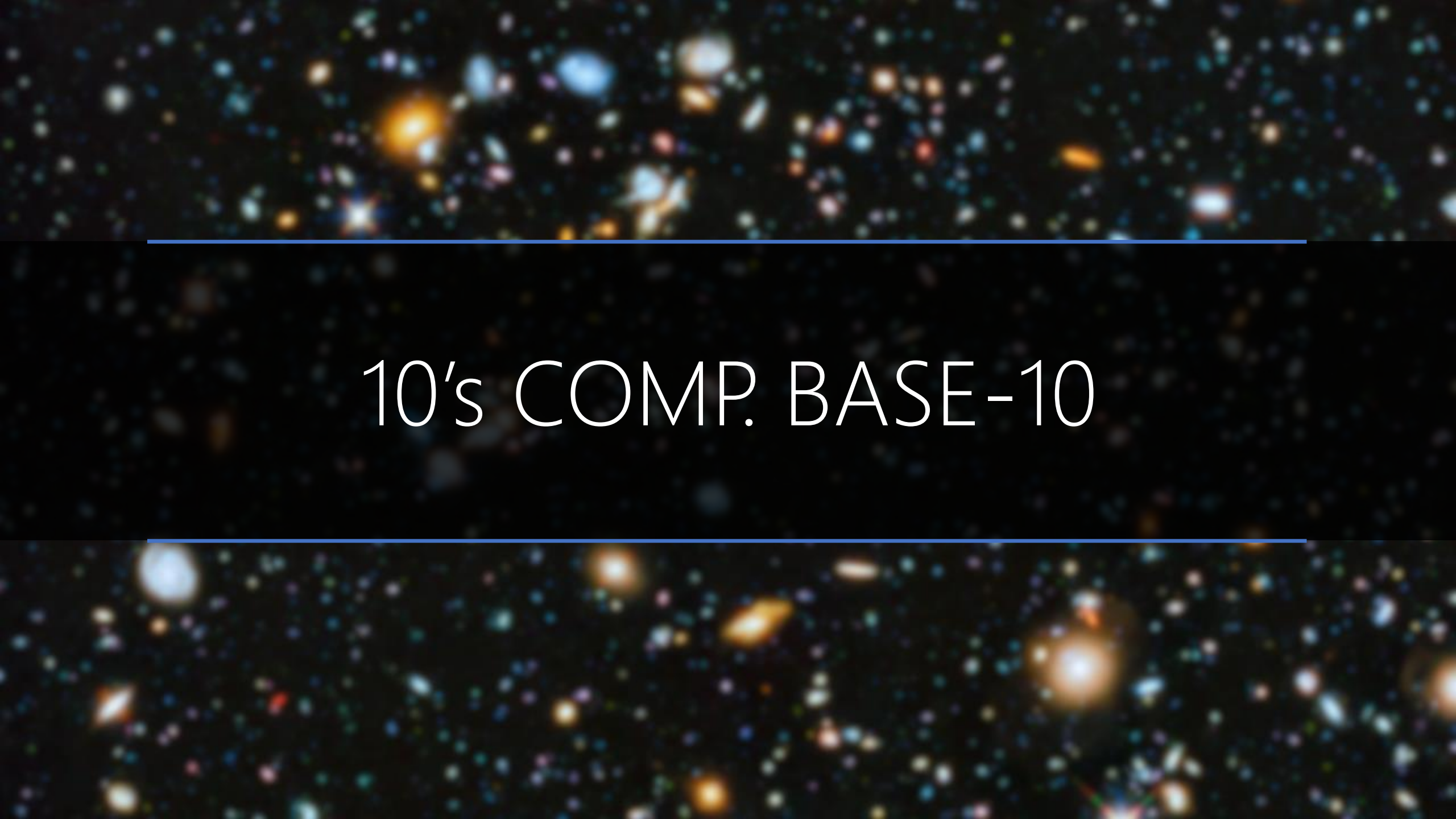
A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in a variety of hues, including blue, orange, yellow, and white, set against a dark, black background. The galaxies are distributed across the entire frame, with some appearing more prominent than others.

2's COMP. BASE-2

[illegible]

The background of the slide is a deep space image showing a dense field of galaxies. The galaxies are of various colors, including blue, orange, and white, and are scattered across the dark background. A horizontal blue line runs across the middle of the image, just above the text.

4's COMP. BASE-4

A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in shades of blue, orange, and white, set against a dark, star-filled background. Two horizontal blue lines are positioned above and below the central text.

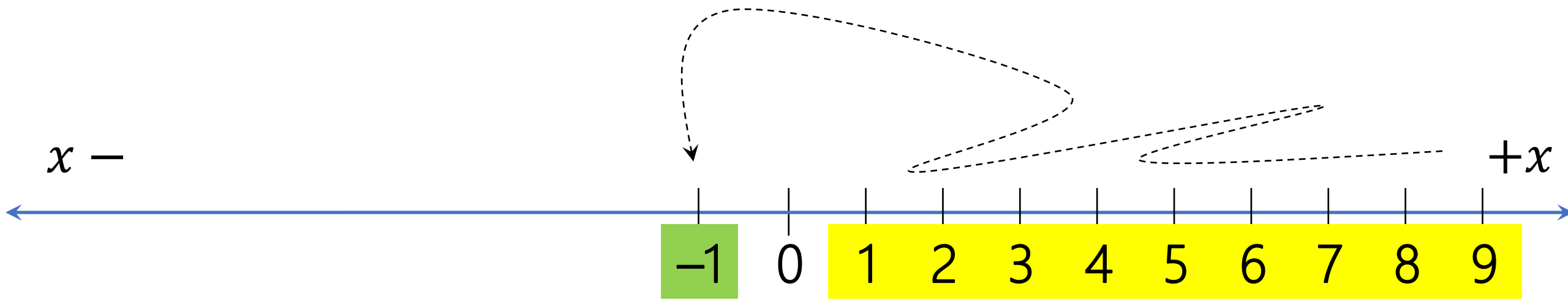
10's COMP. BASE-10

The image features a cosmic background of numerous galaxies in various colors (blue, orange, white) against a black space. A horizontal blue line is positioned above the text, and another is below it.

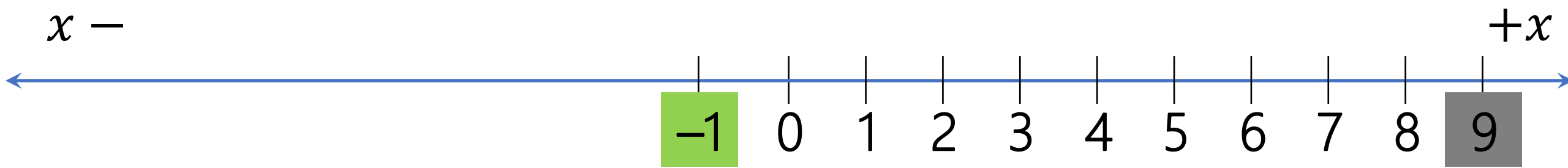
r's COMP. BASE-r

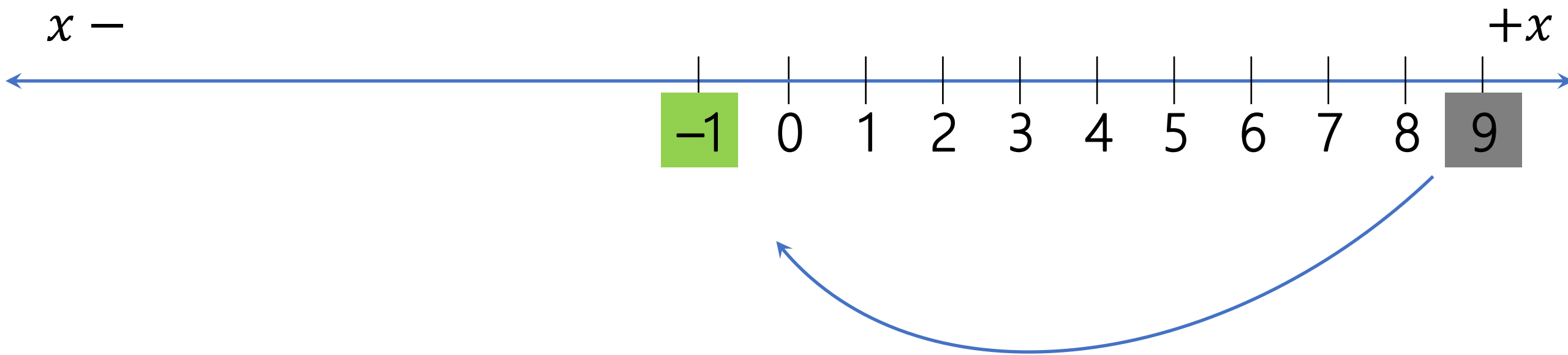
A deep space image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

SIGNED COMPLEMENT

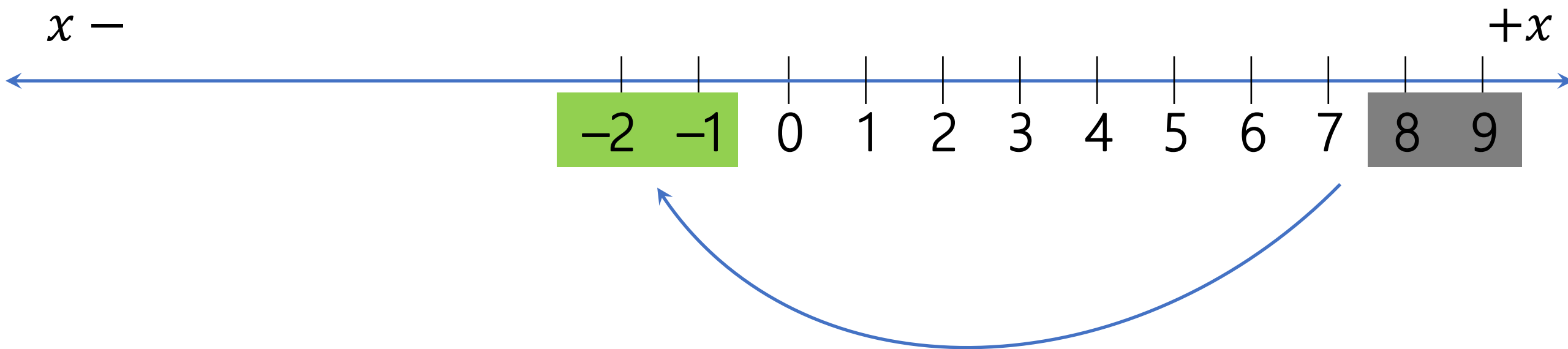


Which one do you scarify?

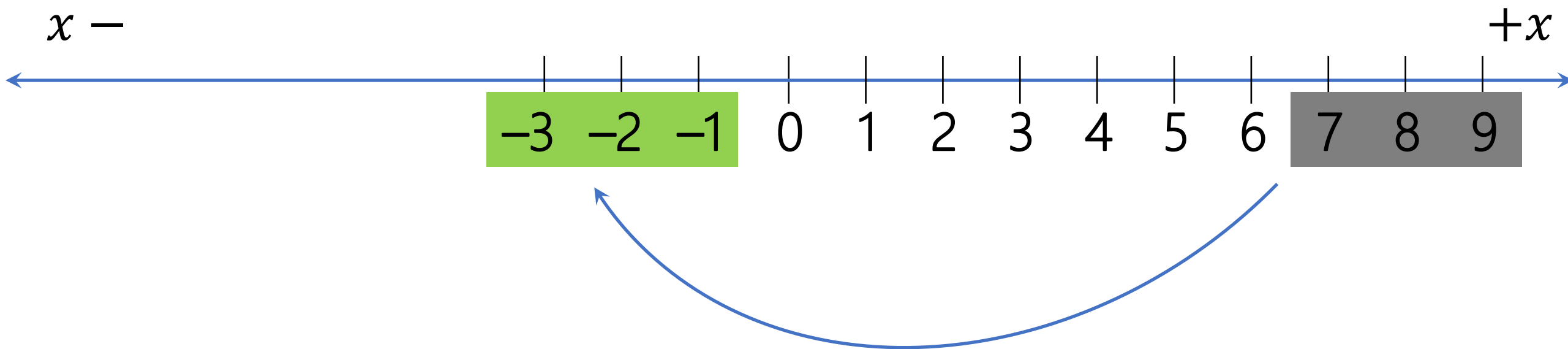




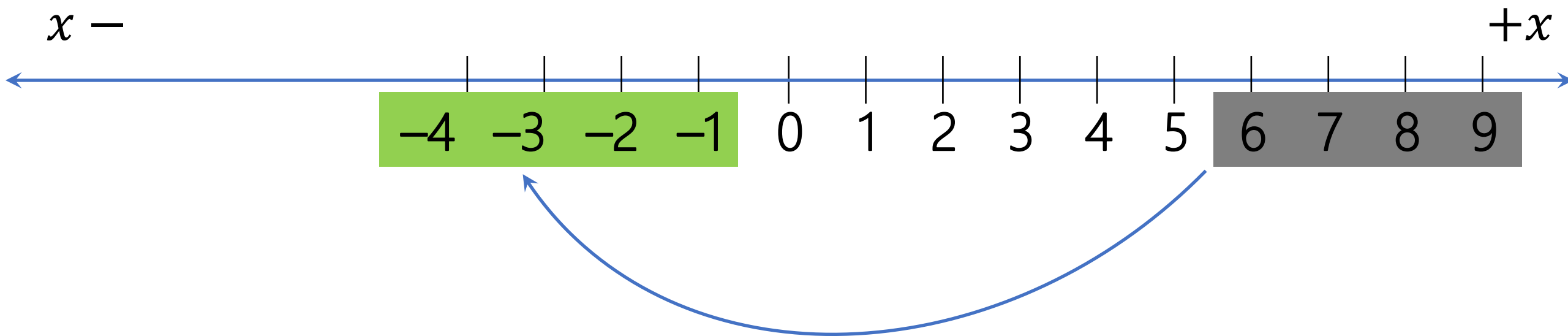
$$\boxed{-1} = -(10^1 - \boxed{9}) = -(10\text{'s comp. } \boxed{9})$$



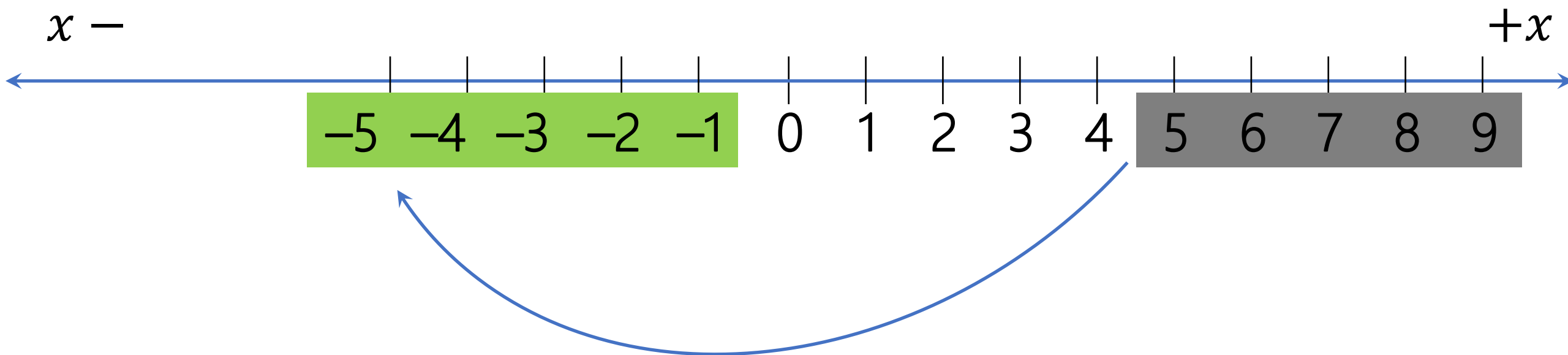
$$\textcolor{red}{-2} = -(10^1 - \textcolor{gray}{8}) = -(10\text{'s comp. } \textcolor{gray}{8})$$



$$\textcolor{red}{-3} = -(10^1 - \textcolor{gray}{7}) = -(10\text{'s comp. } \textcolor{gray}{7})$$

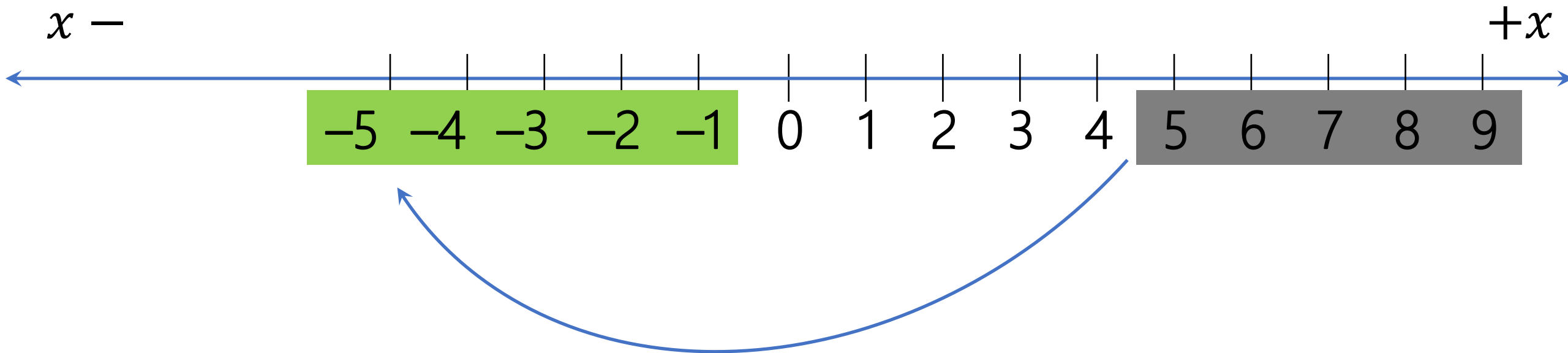


$$\text{[green]} -4 = -(10^1 - \text{[gray]} 6) = -(10\text{'s comp. } \text{[gray]} 6)$$



$$\text{green } -5 = -(10^1 - \text{gray } 5) = -(10\text{'s comp. } \text{gray } 5)$$

Which one you keep/drop, 5 or -5?



$$\text{green } -5 = -(10^1 - \text{gray } 5) = -(10\text{'s comp. } 5)$$

10^0	$-(10\text{'s comp.})$	10^0
0	→	0
1		-9
2		-8
3		-7
4		-6
5		-5
6		-4
7		-3
8		-2
9		-1

10^0	$-(10\text{'s comp.})$	10^0
0	←	0
1		−9
2		−8
3		−7
4		−6
5		−5
6		−4
7		−3
8		−2
9		−1

10^0		10^0
0	Eqaul Share Base-10 Signed 10's comp.	0
1		−9
2		−8
3		−7
4		−6
5		−5
6		−4
7		−3
8		−2
9		−1

10^0		10^0
0	Base-10 Signed 10's comp.	see 0 interpret 0
1		see 1 interpret 1
2		see 2 interpret 2
3		see 3 interpret 3
4		see 4 interpret 4
5 → -5		see 5 interpret -5
6 → -4		see 6 interpret -4
7 → -3		see 7 interpret -3
8 → -2		see 8 interpret -2
9 → -1		see 9 interpret -1



0:03 / 1:26

https://www.youtube.com/watch?v=CjbZF5_4xW8



A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in a variety of hues, including blue, orange, yellow, and white, set against a dark, star-filled background. Two thin, horizontal blue lines are positioned above and below the central text.

SIGNED 10's COMP. Base-10

Signed Radix Complement Base-10

 10^{n-1} 10^{n-2} 10^{n-3}

...

 10^2 10^1 10^0 $0 \leq$

Positive Numbers

 $\leq (10^n - 1) \div 2$

Nothing to do!
Surface == Meaning

Signed Radix Complement Base-10

 10^{n-1} 10^{n-2} 10^{n-3}

...

 10^2 10^1 10^0 $(10^n - 1) \div 2 + 1 \leq$ Negative Numbers $\leq (10^n - 1)$

Although we see positive numbers, we must interpret them negative!

How?

Surface $\neq$ Meaning

Signed Radix Complement Base-10

 10^{n-1} 10^{n-2} 10^{n-3}

...

 10^2 10^1 10^0

$(10^n - 1) \div 2 + 1 \leq$ Negative Numbers $\leq (10^n - 1)$

— (10's comp. of the number we see)

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$n = 7$$

$0 \leq$ Positive Numbers $\leq (10^7 - 1) \div 2$

$(10^7 - 1) \div 2 + 1 \leq$ Negative Numbers $\leq (10^7 - 1)$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$n = 7$$

$$(10^7 - 1) \div 2 = 9,999,999 \div 2 = 4,999,999$$

$$5,805,074 > 4,999,999$$

This number must be interpreted negative!

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$= - (10\text{'s comp. } (5,805,074))$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$\begin{aligned} &= - (10\text{'s comp. } (5,805,074)) \\ &= - (9\text{'s comp. } (5,805,074) + 1) \end{aligned}$$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$\begin{aligned} &= - (10\text{'s comp. } (5,805,074)) \\ &= - (9\text{'s comp. } (5,805,074) + 1) \\ &= - (9,999,999 - 5,805,074 + 1) \end{aligned}$$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$\begin{aligned} &= - (10\text{'s comp. } (5,805,074)) \\ &= - (9\text{'s comp. } (5,805,074) + 1) \\ &= - (9,999,999 - 5,805,074 + 1) \\ &= - (4,194,925 + 1) \end{aligned}$$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
5	8	0	5	0	7	4

$$\begin{aligned} &= - (10\text{'s comp. } (5,805,074)) \\ &= - (9\text{'s comp. } (5,805,074) + 1) \\ &= - (9,999,999 - 5,805,074 + 1) \\ &= - (4,194,925 + 1) \\ &= - (4,194,926) \end{aligned}$$

Signed Radix Complement Base-10

10^6

10^5

10^4

10^3

10^2

10^1

10^0

X

X

X

X

X

X

X

– 3,450,256

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
X	X	X	X	X	X	X

– 3,450,256

We must find its positive complement to represent it!

How?

Signed Radix Complement Base-10

10^6

10^5

10^4

10^3

10^2

10^1

10^0

– 3,450,256

= 10's comp (3,450,256)

Signed Radix Complement Base-10

10^6

10^5

10^4

10^3

10^2

10^1

10^0

– 3,450,256

= 10's comp (3,450,256)

= 9's comp (3,450,256) + 1

Signed Radix Complement Base-10

10^6

10^5

10^4

10^3

10^2

10^1

10^0

$- 3,450,256$

$= 10\text{'s comp } (3,450,256)$

$= 9\text{'s comp } (3,450,256) + 1$

$= 9,999,999 - 3,450,256 + 1$

Signed Radix Complement Base-10

10^6

10^5

10^4

10^3

10^2

10^1

10^0

$- 3,450,256$

$= 10\text{'s comp } (3,450,256)$

$= 9\text{'s comp } (3,450,256) + 1$

$= 9,999,999 - 3,450,256 + 1$

$= 6,549,743 + 1$

Signed Radix Complement Base-10

10^6	10^5	10^4	10^3	10^2	10^1	10^0
6	5	4	9	7	4	4

$$\begin{aligned} & - 3,450,256 \\ & = 10\text{'s comp } (3,450,256) \\ & = 9\text{'s comp } (3,450,256) + 1 \\ & = 9,999,999 - 3,450,256 + 1 \\ & = 6,549,743 + 1 \\ & = 6,549,744 \end{aligned}$$

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black sky. Two horizontal blue lines frame the central text.

SIGNED 16's COMP. Base-16

Base-16	1	B	F	2	B	4
	Is this a negative number or positive?					

Base-16	1	B	F	2	B	4
	How many positions? $n = 6$					

Base-16

1

B

F

2

B

4

What is the maximum positive number?

$$(16^6 - 1) \div 2 = (8,388,607)_{10} = (7FF,FFF)_{16}$$

Base-16

1

B

F

2

B

4

This number is less than $(7FF,FFF)_{16}$
 $(+1BF,2B4)_{16}$

Base-16	E	B	F	2	B	4
	Is this a negative number or positive?					

Base-16	E	B	F	2	B	4
	How many positions? $n = 6$					

Base-16

E

B

F

2

B

4

What is the maximum positive number?

$$(16^6 - 1) \div 2 = (8,388,608)_{10} = (7FF,FFF)_{16}$$

Base-16

E

B

F

2

B

4

This number is greater than $(7FF, FFF)_{16}$
The number is negative, but what is the number?

Base-16	E	B	F	2	B	4
	= − (16's comp. (EBF,2B4))					

Base-16

E

B

F

2

B

4

$$\begin{aligned} &= - (16\text{'s comp. (EBF,2B4)}) \\ &= - (15\text{'s comp. (EBF,2B4) + 1)} \end{aligned}$$

Base-16

E

B

F

2

B

4

$$\begin{aligned} &= - (16's \text{ comp. } (EBF,2B4)) \\ &= - (15's \text{ comp. } (EBF,2B4) + 1) \\ &= - (FFF,FFF - EBF2B4 + 1) \end{aligned}$$

Base-16

E

B

F

2

B

4

$$\begin{aligned} &= - (16\text{'s comp. (EBF,2B4)}) \\ &= - (15\text{'s comp. (EBF,2B4) + 1}) \\ &= - (FFF,FFF - EBF2B4 + 1) \\ &= - (140,D4B + 1) \end{aligned}$$

Base-16

E

B

F

2

B

4

$$\begin{aligned} &= - (16\text{'s comp. (EBF,2B4)}) \\ &= - (15\text{'s comp. (EBF,2B4) + 1}) \\ &= - (FFF,FFF - EBF2B4 + 1) \\ &= - (140,D4B + 1) \\ &= - (140,D4C) \end{aligned}$$

Base-16	X	X	X	X	X	X
	– (140,D4C)					

Base-16

X

X

X

X

X

X

$$\begin{aligned} & - 140, D4C \\ & = 16\text{'s comp}(140, D4C) \\ & = 15\text{'s comp}(140, D4C) + 1 \\ & = FFF, FFF - 140, D4C + 1 \\ & = EBF, 2B3 + 1 \\ & = \text{EBF, 2B4} \end{aligned}$$

Base-16

E

B

F

2

B

4

– 140,D4C

= 16's comp(140,D4C)

= 15's comp(140,D4C) + 1

= FFF,FFF – 140,D4C + 1

= EBF,2B3 + 1

= EBF,2B4

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED r 's COMP. Base- r

Base-r in Radix Complement

 r^{n-1} r^{n-2} r^{n-3}

...

 r^2 r^1 r^0

$0 \leq \text{Positive Numbers} \leq (r^n - 1) \div 2$

Nothing to do!

Base-2: 0,111,...,111

Base-4: 1,333,...,333

Base-8: 3,777,...,777

Base-10: 4,999,...,999

Base-16: 7,FFF,...,FFF

Base-r in Radix Complement

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
$(r^n - 1) \div 2 + 1 \leq$			Negative Numbers		$\leq (r^n - 1)$	
Base-2: 1,000,...,000						
Base-4: 2,000,...,000						
Base-8: 4,000,...,000						
Base-10: 5,000,...,000						
Base-16: 8,000,...,000						

We see positive number, but we interpret negative!
 $= - (r\text{'s comp. } (\#)) = - ((r-1)\text{'s comp. } (\#) + 1)$

Base-r in Radix Complement

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
$(r^n - 1) \div 2 + 1 \leq$		Negative Numbers		$\leq (r^n - 1)$		
Base-2: 1,000,...,000		The r's comp of these borderline numbers are themselves! But in the second have → Negative numbers				
Base-4: 2,000,...,000						
Base-8: 4,000,...,000						
Base-10: 5,000,...,000						
Base-16: 8,000,...,000						

We see positive number, but we interpret negative!
 $= - (r\text{'s comp. } (\#)) = - ((r-1)\text{'s comp. } (\#) + 1)$

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots in various shapes and sizes, including spirals, ellipticals, and irregular forms, set against a dark, star-filled sky. Two thin, horizontal blue lines are positioned above and below the central text.

Practice Base-2,3,4
1, 2, 3, 4 Positions

SIGNED 2'S COMPLEMENT BASE-2

6. Show that in 2's complement binary system, the highest significant position acts like a sign (not the same) as in signed-magnitude binary system. Is this true for any radix-r number system? Justify your answer.

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

SIGNED r 'S COMPLEMENT ARITHMETIC

Signed Magnitude

0	X	+	0	Y	=	0	X+Y
0	X	+	1	Y	=	X<Y (if borrow)	X-Y (if borrow, apply it)
1	X	+	1	Y	=	1	-(X+Y) ≈ X+Y
0	X	-	0	Y	=	X<Y (if borrow)	X-Y (if borrow, apply it)
0	X	-	1	Y	=	0	X=Y
1	X	-	1	Y	=	X>Y (if borrow)	X=Y -X+Y=Y-X (if borrow, apply it)

$$\boxed{X} + \boxed{Y} = \boxed{X+Y}$$

$$\boxed{X} - \boxed{Y} = \boxed{X + (r's \text{ comp } Y)}$$

Last Carry → Ignore

Signed Magnitude

0	X	+	0	Y	=	0	X+Y
0	X	+	1	Y	=	X<Y (if borrow)	X-Y (if borrow, apply it)
1	X	+	1	Y	=	1	-(X+Y) = X+Y

0	X	-	0	Y	=	X<Y (if borrow)	X-Y (if borrow, apply it)
0	X	-	1	Y	=	0	X=Y
1	X	-	1	Y	=	X>Y (if borrow)	X+Y = Y-X (if borrow, apply it)

$$\boxed{X} + \boxed{Y} = \boxed{X+Y}$$

$$\boxed{X} - \boxed{Y} = \boxed{X+(r's \text{ comp } Y)}$$

One adder for both addition and subtraction!

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

SIGNED r 'S COMPLEMENT ADDITION

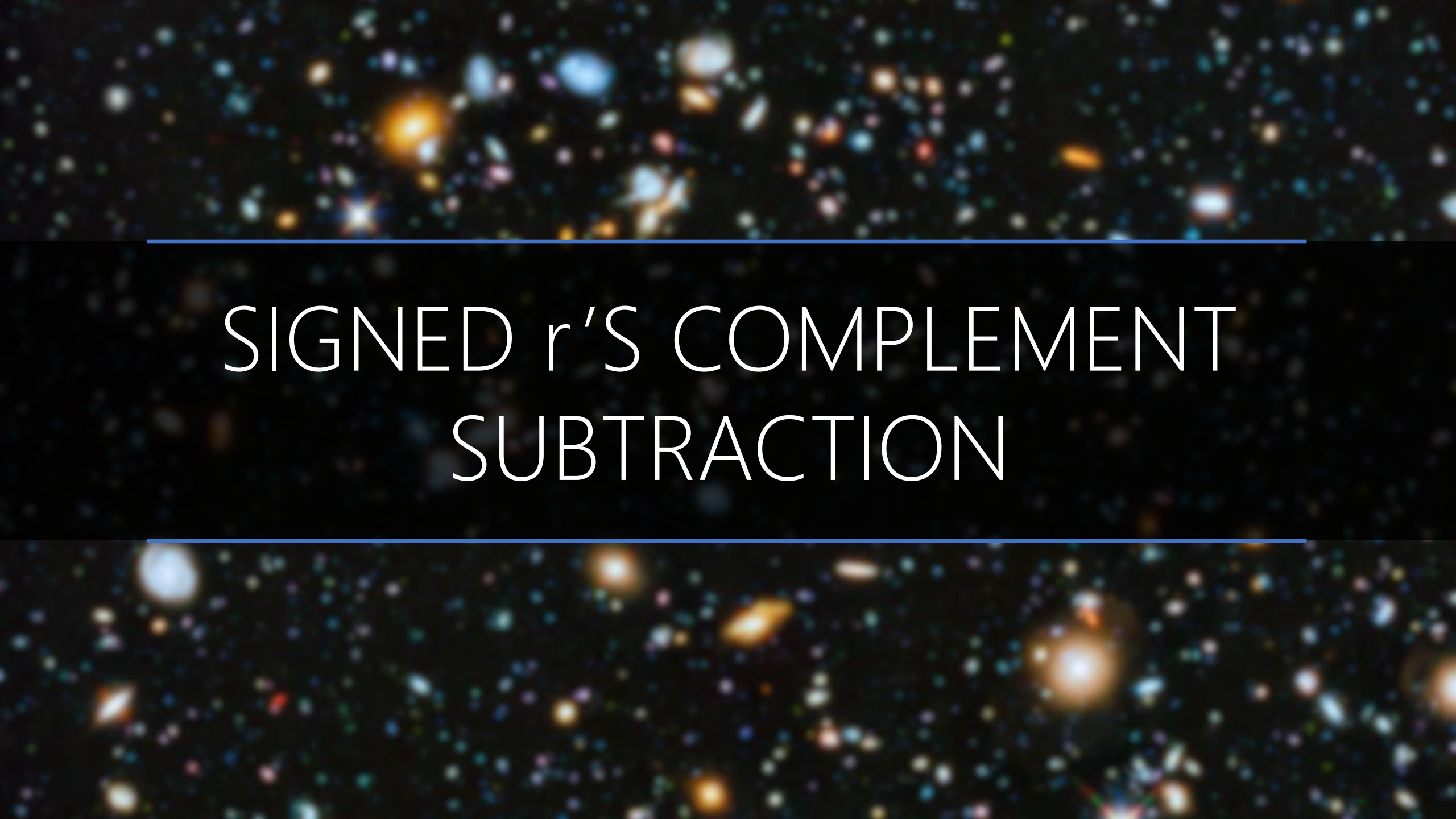
+ Base-2		1	0	1	0	1	1
		0	0	1	1	1	0

+ Base-2		1	0	1	0	1	1
		$\begin{aligned} &> (2^6-1) \div 2 = (011,111)_2 \\ &= -(2's \text{ comp}(\#)) \\ &= -(010101)_2 \\ &= -(21)_{10} \end{aligned}$					

+ Base-2							
	0	0	1	1	1	0	
	$< (2^6 - 1) \div 2 = (011, 111)_2$ $= + (001110)_2$						

			1	1	1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		1	1	1	0	0	1

			1	1	1		
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		1	1	1	0	0	1
		$ \begin{aligned} &> (2^6 - 1) \div 2 = (011, 111)_{10} \\ &= -(2's \text{ comp}(\#)) \\ &= -(000111)_2 \\ &= -(7)_{10} \end{aligned} $					

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies are in various stages of evolution and are colored based on their redshift, with blue representing higher redshift (further away) and yellow/orange representing lower redshift (closer). The galaxies are scattered across the dark cosmic void, with some appearing as bright, distinct shapes and others as faint, distant points of light.

SIGNED r 'S COMPLEMENT SUBTRACTION

– Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(8)_{10}$	0	0	1	0	0	0

Base-2	— $-(21)_{10}$	1	0	1	0	1	1
	$+(8)_{10}$	0	0	1	0	0	0
	We must convert the subtraction to addition. Eq., addition with $-(8)_{10}$!						

Base-2	 $-(21)_{10}$	1	0	1	0	1	1
	$+(8)_{10}$	0	0	1	0	0	0
	$= -(2's \text{ comp. } (001000))$ $= -(1's \text{ comp}(001000) + 1)$ $= -(\text{NOT}(001000) + 1)$ $= -(110111 + 1)$ $= - (111000)$						

+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0

Last Carry → Ignore

+ Base-2	1	1	1				
	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0
		1	0	0	0	1	1

		1	1				
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0
		1	0	0	0	1	1
		Done! This is the Result.					

		1	1				
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(8)_{10}$	1	1	1	0	0	0
	$-(29)_{10}$	1	0	0	0	1	1

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

SIGNED r 'S COMPLEMENT OVERFLOW

$$\boxed{X} + \boxed{Y} = \boxed{X+Y}$$

The + result of two **negative** numbers → **positive**

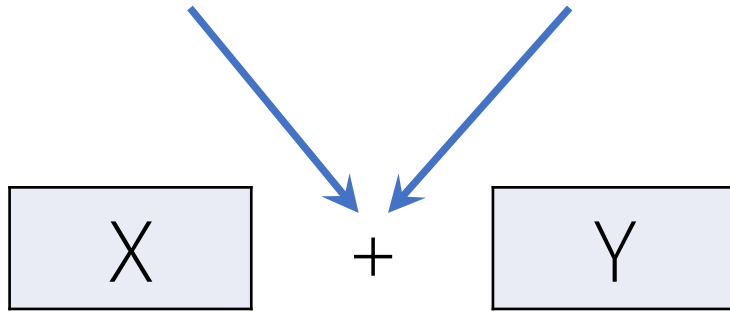
The + result of two **positive** numbers → **negative**

$$\boxed{X} - \boxed{Y} = \boxed{X + (\text{r's comp } Y)}$$

The + result of two negative numbers → positive
The + result of two positive numbers → negative

Originally Add

After Comp. for Subtraction



The + result of two **negative** numbers → **positive**

The + result of two **positive** numbers → **negative**

Always Check!

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

OVERFLOW

Example

– Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0

Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
	We must convert the subtraction to addition. Eq., addition with $-(14)_{10}$!						

Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
	$= (2's \text{ comp. } (001110))$ $= (1's \text{ comp}(001110) + 1)$ $= (\text{NOT}(001110) + 1)$ $= (110001 + 1)$ $= (110010)$						

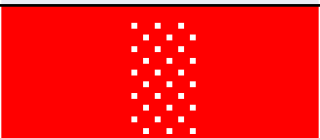
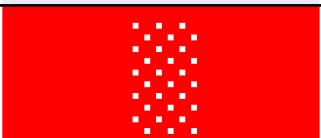
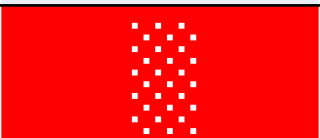
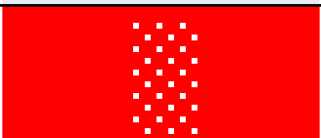
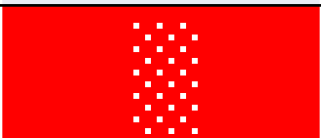
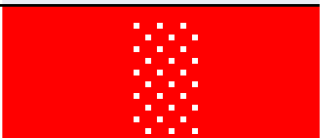
+ Base-2	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0

Two Negative Numbers ...

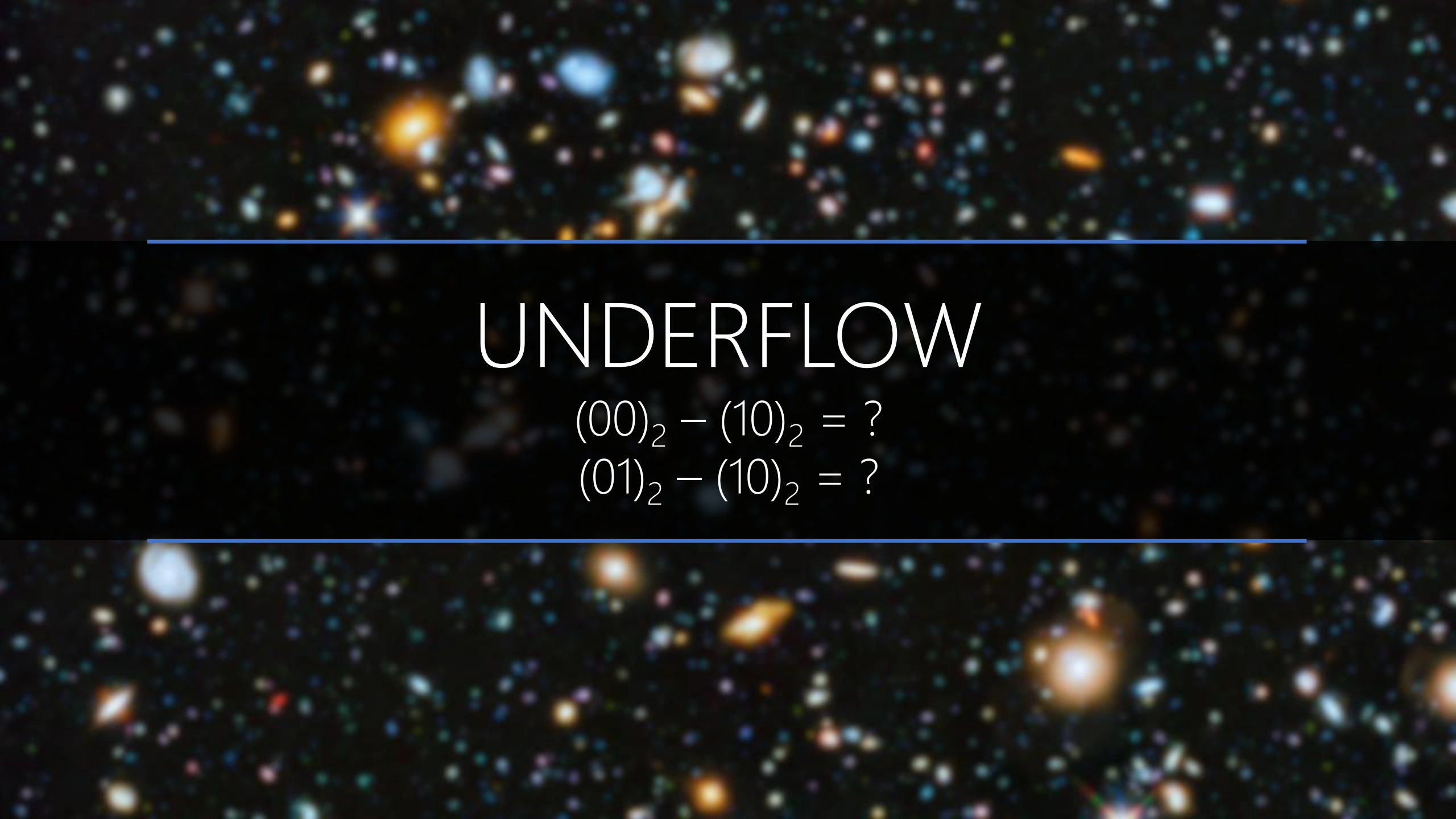
Last Carry → Ignore

+ Base-2	1						
	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0
		0	1	1	1	0	1

					1		
+	—(21) ₁₀	1	0	1	0	1	1
Base-2	—(14) ₁₀	1	1	0	0	1	0
		0	1	1	1	0	1
		$< 2^6 - 1 \div 2 = (011, 111)_{10}$ $= (011101)_2$ Positive Number					

Base-2	+				1		
	$-(21)_{10}$	1	0	1	0	1	1
	$-(14)_{10}$	1	1	0	0	1	0
							
		Overflow! Don't rely on the result!					

Sum of two negative numbers become positive!

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UNDERFLOW

$$(00)_2 - (10)_2 = ?$$

$$(01)_2 - (10)_2 = ?$$